

Qiren Liu

AI Agent / Backend Software Engineer

Phone: 13382183295 | Email: lqr_ld@163.com

Professional Summary

Software engineer with 4 years of experience at Microsoft Advertising, focused on **AI agents and advertising backend systems**. Designed and built an evidence-first retrieval agent for multi-repository change history and regression investigations, with end-to-end experience in hybrid retrieval, agent orchestration, MCP, abstention gates, layered evaluation, and observability. Also delivered backend systems for image asset governance, Deal synchronization, and asynchronous video recommendations, as well as large-scale Campaign UI features.

Core Skills

- AI / Agents:** Agentic RAG, Embeddings, FTS/Dense/Exact Hybrid Retrieval, RRF, Evidence Grounding, Agent Evaluation, MCP, Prompt/Context Engineering, SSE
- Backend:** C#, Python, Cosmos DB
- Frontend:** JavaScript, React, Fluent UI
- Data & Engineering:** sqlite-vec, Azure Blob/Table Storage, Kubernetes, Kusto/KQL, Azure DevOps CI/CD

Education

- University of Electronic Science and Technology of China | M.S., School of Network Engineering** | 2019.09 - 2022.06
- University of Electronic Science and Technology of China | B.E., School of Information and Communication Engineering** | 2015.09 - 2019.06

Professional Experience

Microsoft | Software Engineer, Microsoft Advertising / Microsoft AI

2022.08 - 2026.08

Core AI Agent Project: Commit AI Resolver | Change History Retrieval & Regression Investigation Agent

JavaScript / SQLite FTS5 / sqlite-vec / RRF / MCP

- Designed a bounded **Intent Extractor -> Hybrid Retriever -> Evidence Gate -> Answer Synthesizer -> Answer Evaluator** investigation pipeline for multi-repository, long-horizon commit search and production regression diagnosis. Used precomputed history indexes as a low-cost candidate generation layer instead of relying on the model to read an entire time window.
- Built both an internal multi-repository system and a public reproduction: daily JSON as the auditable source of truth, with SQLite metadata/FTS5 and sqlite-vec as rebuildable derived indexes. Enriched 27,646 React commits with changed files, affected areas, and complete messages, while versioning the embedding model, 1,024 dimensions, query instructions, document templates, and index format as a compatibility contract.
- Implemented **Direct SHA + FTS5 + Dense + Multi-query** retrieval: direct hits for exact identifiers, structured pre-filters, and separate searches for semantic rewrites and ticket titles. Combined heterogeneous rankings with weighted RRF and retained channel-level contributions for diagnosis.
- Added a deterministic Evidence Gate before generation, combining exact hits, explicit metadata constraints, dense scores, and cross-channel consistency to return SEARCH / ABSTAIN / ASK_USER. This prevented nearest-neighbor results from being treated as sufficient evidence; the answer layer also validated citations and commit authenticity.
- Implemented Evaluator-driven bounded retries for up to three rounds, with keyword/date-window adjustments, stale-retry detection via result ID sets, and best-so-far retention. Exposed six MCP tools over Streamable HTTP so IDE agents could reuse retrieval, diff, and summarization capabilities.

- Isolated dynamic business content from the System Prompt and introduced strict JSON Schemas for four agent types, deterministic validation of candidate commits/URLs, and compatible fallbacks, reducing prompt injection, format drift, and evidence hallucination risks.
- Established a Prompt Registry, deterministic stable/candidate traffic splitting, a kill switch, and automatic rollback after consecutive failures. Combined a 10-case Golden Eval, CI gates, and per-agent token/latency/error telemetry in SQLite into a prompt quality loop, with 185 automated tests passing.
- Built a versioned 75-case engineering regression harness and created and fully evaluated 461 RCA gold cases from real GitHub Issue -> closing PR -> fix commit chains. Using issue-lifecycle time windows and a locally trained learning-to-rank model, improved Recall@20 on a grouped 134-case held-out test set from 70.90% to 94.78%, while preserving per-case traces, manual review, and explicit release-gate boundaries.

Advertising UI Projects

- Delivered 490+ production merges in the large-scale Microsoft Advertising Campaign UI. Reduced P95 latency on key pages by **40%-72%** through systematic performance optimization and shipped image asset quality prediction, third-party tracking, Media Deal, and ad recommendation features.

Advertising Backend Projects

C#/ .NET / WCF / SOAP / Bulk API / Task Engine / SQL Server / Azure Queue / Cosmos DB/ADLS

Image Asset Lifecycle Governance

- Designed and delivered an end-to-end workflow for identifying, cleaning up, and restoring orphaned image assets, using data-layer multi-account cleanup/activation stored procedures and a middle-tier Kubernetes CronJob for batch governance.
- Built a WCF email notification service and templates for affected accounts, with error handling, failure alerts, and multi-environment configuration to ensure reliable operation.
- Added image validation and functional tests for ad extensions and Smart Pages, resolved false-positive issues, and migrated OSManagementDAM capabilities to the middle tier for centralized maintenance.

Programmatic Deal Synchronization & Delivery Governance

- Led the automated synchronization of Xandr Deals to MS Deals from inception, delivering paginated API ingestion, model mapping and validation, consistent behavior across SOAP/Bulk/WCF entry points, targeting constraints, market-by-market staged GA, and safe retirement of the legacy version.

Video Recommendation & Creative Generation Pipeline

- Built a configuration-driven daily batch pipeline that selected a target number of ads, read database mapping data from Cosmos DB/ADLS, and generated recommended videos. Ensured reliable calls to an external generation API through Azure Queue rate limiting, allow/deny lists, batch-level degradation, and partial-success handling.